

Dev. Work Report

Project: Sales Analysis Pipeline – TechStore

Goal

Document technical errors, debugging strategies, repository organization, deployment workflow, and lessons learned during the development and publication process.

STEP 1 — Repository Configuration

Problem

The repository could not be found during remote connection attempts.

Cause

Incorrect repository URL configuration caused routing failure.

Solution

Corrected the repository name directly in the GitHub interface and validated the remote configuration.

Learning

- Importance of repository validation
- URL structure verification
- Remote connection workflow

Improvement

- Validate repository configuration before authentication setup
- Standardize repository naming conventions

STEP 2 — Authentication and Access

Problem

Authentication failed even with a valid GitHub token.

Cause

The token was automatically invalidated after exposure during the workflow process.

Solution

Migrated authentication strategy from HTTPS token access to SSH key authentication.

Learning

- Authentication security practices
- SSH workflow configuration
- Credential management importance

Improvement

- Avoid token exposure during configuration
- Use SSH authentication as default workflow

STEP 3 — SSH Permission Configuration

Problem

SSH connection failed during repository access.

Cause

The public SSH key had not been registered in the GitHub account settings.

Solution

Generated and linked the public SSH key to the GitHub account manually.

Learning

- SSH key structure and validation
- Access permission management
- Secure remote authentication workflow

Improvement

- Configure SSH authentication before repository operations

STEP 4 — Directory Structure and Tracking

Problem

Git could not locate the project files during tracking operations.

Cause

The terminal targeted the incorrect directory level while the actual project files were stored inside the src/ folder.

Solution

Adjusted the Git tracking scope to the correct project directory structure.

Learning

- Directory hierarchy organization
- Git tracking workflow
- Importance of project structure consistency

Improvement

- Organize folder hierarchy before version control initialization
- Create standardized project structures

STEP 5 — Cache and Synchronization Failure

Problem

GitHub Desktop remained stuck during synchronization attempts.

Cause

Corrupted cache references and fragmented commit history affected repository synchronization.

Solution

Applied a manual deployment workflow:

- Downloaded clean repository ZIP
- Reorganized project structure locally
- Uploaded files directly through GitHub web interface

Learning

- Alternative deployment strategies
- Manual repository recovery workflow
- Cache failure troubleshooting

Improvement

- Maintain clean commit history
- Validate synchronization state frequently

Workflow Structure

The final project pipeline followed this structure:

- Data Extraction and Reading
- Data Cleaning and Grouping
- Statistical Visualization

- Automated Executive Report Generation in Excel

Technical Reflection

The debugging and deployment process reinforced:

- workflow organization
- troubleshooting methodology
- repository management
- persistence during technical failures
- adaptive problem-solving strategies

The project also improved understanding of:

- Git workflows
- authentication systems
- data processing pipelines
- debugging logic
- deployment organization

Author: Géssica Nascimento
Systems Analysis & Development Student

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